

Dynamics & flexibility

Week 37 — why good firms borrow less than they can

Advanced Corporate Finance (EBM232A05)

University of Groningen · Sebastian Pfeil

Week 36 left two loose ends about *leverage*:

- Lemmon et al. (2008): most of it is a stable, firm-specific “something”.
- Korteweg (2010): firms sit **below** the optimum, and adjust **sluggishly**.

Wk	Friction	Effect	Evidence
36	Personal taxes (Miller)	value $\uparrow$ / $\downarrow$	net benefit (Korteweg, 2010)
37	Dynamics / flexibility	? today	—

Today: the first friction that makes “too conservative” the **right** answer — by adding *time*.

- The puzzle: managers rank **flexibility** above the tax shield — survey evidence.
- The benchmark you own: the **static** trade-off and its optimum $\bar{D}$.
- The twist: add a tomorrow — **debt capacity as a real option**; lever up only when $\tau I \geq pG$.
- A second route to the same place: **debt overhang** (Myers, 1977).
- The upshot: the comparative statics *are* the cross-section (growth firms borrow less).
- Why firms are also **slow** to adjust — and the Week-37 ledger row.

One friction — **time** — two mechanisms, one lesson: conservative leverage can be optimal.

The puzzle: money left on the table

What managers say they are doing

Graham and Harvey (2001) put the capital-structure question to ~400 CFOs; Graham (2022) revisits it. The headline is stable:

- **Financial flexibility** is the *most* important determinant of debt policy — ranked above the tax shield.
- Firms deliberately **preserve borrowing capacity** and defend a target credit rating.
- The textbook trade-off factors (taxes, distress) rank *below* “keeping our options open”.

A survey, not a proof — but a striking one: the people who choose leverage say the *static* trade-off is not what drives them.

The survey, up close

Graham (2022), on how firms *actually* manage leverage:

- About **half** report a target debt ratio — so the trade-off is alive — but as a *range*, not a point.
- Day to day they watch a **Debt/EBITDA** rule of thumb — what **rating agencies** and **covenants** track, i.e. distance to a downgrade.
- **Flexibility** means one concrete thing: keep the **capacity to raise more debt** when it is needed.
- Low interest rates and lower corporate tax rates have *shrunk* the tax benefit — yet leverage did not jump.

Targets, pecking order, rules of thumb — they coexist. The common thread, and today's model, is **preserving capacity**.

The shield they leave on the table

The static trade-off says: borrow until the marginal tax shield equals the marginal distress cost — then **stop at $\bar{D}$** . It predicts firms *exhaust* the useful shield.

- Korteweg (2010): the median firm sits **below** that optimum.
- Graham and Harvey (2001): firms do so **deliberately**.

So what is the forgone tax shield **buying**? The static theory has no answer — it has no *tomorrow*. So let us add one.

The benchmark you own

The static trade-off you own (Myers, 1984)

From the bachelor: net tax benefit (τ per dollar — the *net*, post-clawback shield) against expected distress cost.

$$\max_D \underbrace{\tau D}_{\text{tax shield}} - \underbrace{c(D)}_{\text{expected distress}}, \quad c' > 0, c'' > 0 \Rightarrow \text{interior } \bar{D}.$$

- Marginal shield *flattens* (non-debt tax shields; useless if **unprofitable**); distress cost is **convex**. Their crossing pins down $\bar{D}$.
- But it is **one-shot**: choose D once and live with it — no future, no flexibility.

$\bar{D}$ is the firm's **debt capacity** (Myers, 1977; Myers, 1984). The dynamic question: go *all the way* there?

The twist: debt capacity as a real option

One new ingredient: a tomorrow

Keep everything from the static trade-off; add a second date.

- $t=0$: choose debt $D_0 \in [0, \bar{D}]$. Net tax shield τD_0 ; capacity ceiling $\bar{D}$ (the static optimum).
- $t=1$: with probability p , an investment opportunity arrives — cost I , NPV $G > 0$ — that can only be debt-financed. (External equity is prohibitively costly — the friction; more in Week 38.)
- The firm can fund it **only if** it kept room: $\bar{D} - D_0 \geq I$.

What is left out — ask it of every model: no partial funding, one opportunity, equity shut off. Each is a dial we could turn; none changes the punchline.

The trade-off, now dynamic

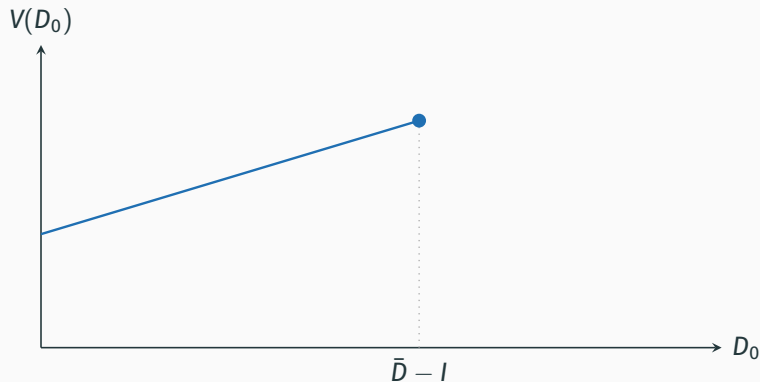
Firm value as a function of today's debt (the indicator $\mathbf{1}\{\cdot\}$ is 1 when there is room to fund, else 0):

$$V(D_0) = \underbrace{\tau D_0}_{\text{tax shield today}} + \underbrace{p G \cdot \mathbf{1}\{D_0 \leq \bar{D} - I\}}_{\text{value of staying fundable}}.$$

- Raising D_0 buys more *certain* shield — but past $\bar{D} - I$ it **kills** the ability to fund the opportunity.
- So only two choices can be optimal: **preserve** ($D_0 = \bar{D} - I$) or **lever to capacity** ($D_0 = \bar{D}$).

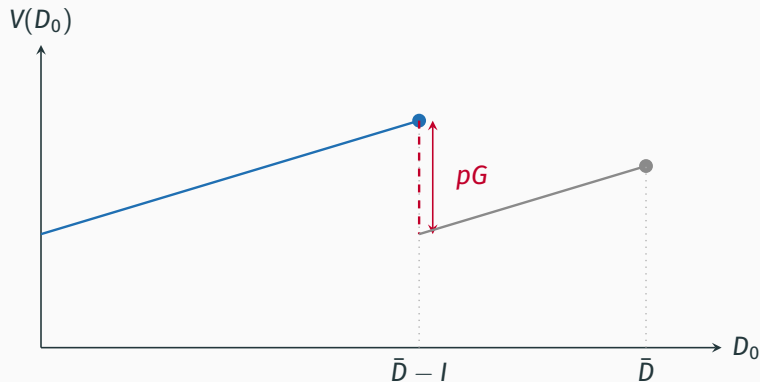
The whole story is the jump at $D_0 = \bar{D} - I$: the last slice of capacity is worth its tax shield *plus* the option it destroys.

The same trade-off, as a picture



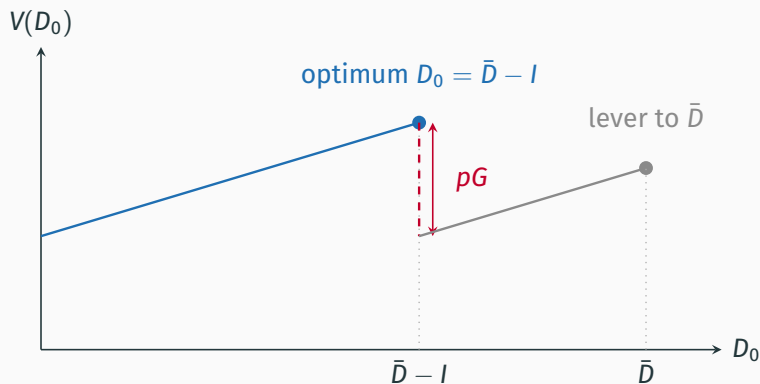
While the firm still has room for tomorrow, every extra euro of debt simply earns the shield: value climbs at **slope τ** .

The same trade-off, as a picture



At $\bar{D} - I$ the room runs out. The next euro of debt **destroys the option**: value drops by pG , then climbs again at slope τ — from lower down.

The same trade-off, as a picture



Two candidate peaks, so compare them. With $pG > \tau I$ the higher one is at $D_0 = \bar{D} - I$: the firm stops *short* of $\bar{D}$ – rational under-leverage.

Debt capacity as a real option

Two ways to use the marginal slice of capacity I :

- **Lever up now**: capture the tax shield τI — *certain*, today.
- **Preserve it**: keep the option to raise I at $t=1$ and fund the opportunity — worth pG in expectation.

Go to the static maximum only if the certain shield beats the option:

$$\text{lever to capacity} \iff \underbrace{\tau I}_{\text{tax shield, sure}} \geq \underbrace{pG}_{\text{option value of preserved capacity}}$$

Reverse the inequality and the firm **stays below** the static optimum — *rational* under-leverage.

Put in numbers

Net shield $\tau = 20\%$ (small — it is the *post-clawback* shield from last week), capacity $\bar{D} = 100$; at $t=1$, with probability $p = 0.5$, an opportunity needs $I = 60$ of debt and delivers NPV $G = 25$.

Choice at $t=0$	Tax shield	Exp. opportunity	Firm value
Lever to capacity ($D_0 = 100$)	20	0 ↓	20
Preserve ($D_0 = 40$)	8	12.5 ↑	20.5

- Forgo $\tau I = 12$ of *certain* shield to hold a real option worth $pG = 12.5$.
Preserve.
- The firm **under-levers by $I = 60$** : it borrows 40, not 100 — capacity it *could* use, deliberately kept free.

You have seen this option before

Spare debt capacity is a **real option**: the right, not the obligation, to raise cheap finance later.

Lever up today

= **exercise early**. You grab the tax shield (the “dividend”) but you *kill the option*.

Preserve capacity

= **keep the option alive**. You forgo the shield now to keep the flexibility.

The bachelor rule for an American call: exercise early only if the dividend justifies it. Here the “dividend” is τI and the option is worth pG — **same tool, new destination**. (Gamba and Triantis 2008 value this flexibility formally.)

A second route: debt overhang

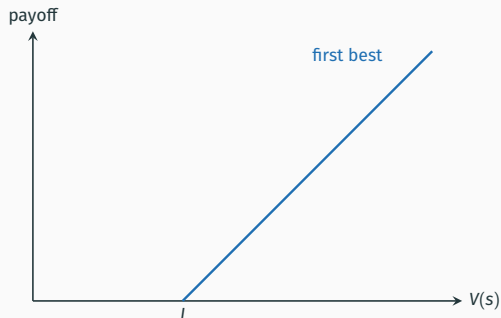
A second route to the same place: debt overhang

Turn it around. A growth firm's value is assets in place *plus* growth options: $V = V_A + V_G$. A growth option is itself a real option — invest I tomorrow, receive $V(s)$ in state s (Myers, 1977).

- **All-equity**: shareholders invest whenever it pays, $V(s) \geq I$ — first best.
- **With risky debt** of face value P already outstanding: shareholders still pay I , but creditors are paid first — so they invest only if $V(s) \geq I + P$.

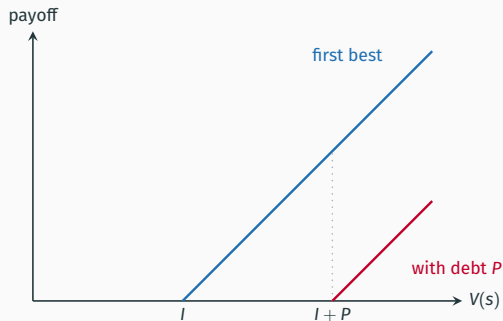
Debt raises the hurdle by P : some **positive-NPV** projects are skipped — **underinvestment**. A firm rich in growth options protects them by **borrowing less**.

Debt overhang, as a picture



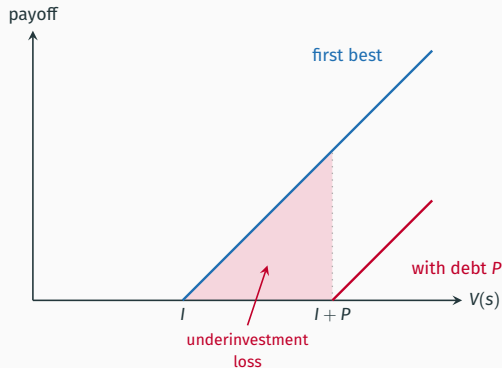
All-equity: shareholders pay I and keep everything above it, so they invest whenever the project pays for itself — the hurdle is $V(s) \geq I$. **First best.**

Debt overhang, as a picture



Now the firm already owes P , and creditors are paid *first*. Shareholders still pay the whole I , so they invest only if $V(s) \geq I + P$: the hurdle **moves right**.

Debt overhang, as a picture



Between the hurdles: the **red wedge** of positive-NPV investment shareholders **forgo**. Debt is a **liability for growth firms**, a **discipline** for mature ones (Myers, 1977; Jensen, 1986).

**Comparative statics = the
cross-section**

Who under-levers most?

Read the comparative statics off both mechanisms — $\tau I \geq pG$ and the overhang:

Firm with...	...does what	Leverage
a more likely future need ($p \uparrow$)	preserves capacity	$\downarrow$
a bigger future opportunity ($G \uparrow$)	preserves capacity	$\downarrow$
many growth options ($V_G \uparrow$)	avoids overhang	$\downarrow$
costlier future external finance	preserves capacity	$\downarrow$
a bigger <i>net</i> tax shield ($\tau \uparrow$)	levers up	$\uparrow$
stable, cash-rich, few projects	levers up	$\uparrow$

Growth firms rationally look “under-levered”; mature cash cows carry more debt. More uncertainty only *raises* the option’s value (convexity).

The cross-section agrees

Frank and Goyal (2009) ask which leverage factors are **reliably** important. The robust signs:

Factor	On leverage	Flexibility reading
Market-to-book (growth)	↓	preserve capacity / avoid overhang
Profitability	↓	internal funds first; keep debt dry
Firm size	↑	stable, diversified — less to insure
Tangibility (collateral)	↑	higher $\bar{D}$; more to borrow against
Industry median leverage	↑	shared technology / opportunity set

The most robust cross-sectional fact — **growth firms borrow less** — is exactly the flexibility motive. (Profitability's sign is also the pecking order; Week 38.)

The other half: sluggish adjustment

The other half: why also slow to adjust?

Korteweg's firms are below the optimum *and* adjust **sluggishly**. The option story gives the *level*; **adjustment costs** give the *speed*.

- Firms hold a **target range**, not a point, and tolerate deviations **for years** rather than pay to rebalance (Fischer et al., 1989).
- Much debt is **transitory** — raised for a lumpy investment, then paid down, **restoring capacity** (DeAngelo et al., 2011).

A target *below* the static optimum (option value), approached *slowly* (adjustment costs). That is Korteweg's picture, now with a mechanism.

Ledger & wrap

Conservative leverage is not timidity — it is the firm **holding a real option**. Two mechanisms, one lesson: preserve capacity to **finance** tomorrow's project (flexibility), and avoid the **overhang** that would deter it (Myers, 1977). Both say a firm rich in future opportunities **under-levers**.

Two things to carry forward:

- This **models** what Korteweg (2010) only *measured*: firms sit below the static optimum on purpose — the more valuable its future opportunities, the further below.
- “Identify the friction” now includes **time**: a choice that looks wrong in a one-period world can be right once tomorrow exists.

The frictions ledger — Week 37

Two rows now. Bring this back next week.

Wk	Friction	Effect	Evidence
36	Personal taxes (Miller)	value $\uparrow$ / $\downarrow$	net benefit (Korteweg, 2010)
37	Dynamics / flexibility	leverage $\downarrow$	flexibility survey (Graham and Harvey, 2001)

The recurring question, every lecture: **which friction, and in which direction?**

What is new here — and what is not

You already own:

- the static trade-off, $\bar{D}$, distress costs
- options, real options, early exercise
- Korteweg's "below optimum" as a *fact*

This week adds:

- **dynamics**: capacity as an option
- under-leverage as an **optimum**, not a puzzle
- **overhang** and the growth-debt link

Same machinery — option payoffs, early exercise — pointed at a result the bachelor left as an unexplained fact.

Reading, and this week

- **This week:** Myers (1977) (Sections 1–3 — the debt-overhang model) and Graham and Harvey (2001) / Graham (2022) (the flexibility survey).
- **Evidence to keep:** Frank and Goyal (2009) (reliable factors) and Korteweg (2010) (below the optimum).
- **Tutorial (Wed):** the $\tau I \geq pG$ threshold as a numeric drill; the project's data/WRDS thread continues.
- **Next week:** when *external* finance is costly — and how one firm's constraint becomes the **macro** accelerator (Week 38). The project **scoping memo** is the gate.

Under-leverage is not caution — it is a firm holding an option on its own future.

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